

Reviewer Pack — Minimal Rank of Quantum Group Representations and DPLL Proof...

Entry 03e72474f51b · INCONCLUSIVE

SEC autonomous research engine — attribution: Ludovico Kubler

2026-06-07 05:21:15 UTC

Minimal Rank of Quantum Group Representations and DPLL Proof Tree Depth

Entry ID: 03e72474f51b **Verdict:** INCONCLUSIVE **Recorded:** 2026-06-07 05:21:15 UTC

1. Conjecture

Field A (mathematical branch): Quantum Groups **Field B** (complexity object): Boolean Satisfiability (DPLL Proof Complexity)

Statement:

For any Boolean formula φ with n variables, the minimal rank of a quantum group representation that distinguishes φ from its negation is at most $O(n^{1.5})$. Furthermore, for every instance of φ , the depth of the DPLL proof tree is lower-bounded by the minimal rank of such a quantum group representation.

Rationale (proposer's reasoning):

Quantum groups have been used in algebraic combinatorics and can represent complexity-theoretic objects like polynomials over finite fields. The use of quantum group representations to distinguish between SAT instances may reveal new connections between representation theory and computational complexity, potentially leading to improved proof systems for SAT.

Taxonomy category: QuantumGroups (status at proposal time:)

2. Pre-registration (Popper-style)

Hash (SHA-256 prefix): 90cf6a766178aefc

This hash commits to the conjecture statement, acceptance criterion, and seed list **before** the empirical test runs. Tampering with the test or its data after this hash was computed would be detectable.

Acceptance criterion:

If the minimal rank of quantum group representations distinguishing φ from its negation is $\leq O(n^{1.5})$ and the DPLL proof tree depth for φ is $\geq$ the depth corresponding to this rank, then the conjecture is supported.

3. Barrier filter (F1)

Final: PASS

Barrier	Verdict	Confidence	LLM ₁	LLM ₂
RELATIVIZATION	UNCERTAIN	1.00	UNCERTAIN	HITS
NATURAL_PROOFS	SAFE	0.50	UNCERTAIN	UNCERTAIN
ALGEBRIZATION	UNCERTAIN	0.90	HITS	UNCERTAIN
KARP_LIPTON	SAFE	0.50	UNCERTAIN	UNCERTAIN

4. Novelty audit

Verdict: NOVEL against 0 hits across arXiv + Semantic Scholar + ECCC

Search queries (3): - "quantum groups" AND "DPLL proof complexity" - "minimal rank" AND "quantum group representations" AND "Boolean satisfiability" - "depth of DPLL proof tree" AND "minimal rank of quantum group representations"

5. Empirical test harness

Multi-seed protocol: 5 seeds (11, 23, 37, 53, 71)

Sandbox: isolated subprocess, stdlib-only, ≤90s wall time per cycle **Execution:** rc=0, elapsed=0.2s

5.1 Generated Python source

```
import random
import math

def run_trial(seed: int) -> dict:
    random.seed(seed)

    def generate_boolean_formula(n):
        return ''.join(random.choice('01') for _ in range(2**n))

    def quantum_group_representation(formula):
        # Placeholder for actual quantum group representation logic
        return len(formula) # Simplified example

    def dpll_proof_tree_depth(rank):
        # Placeholder for actual DPLL proof tree depth logic
        return rank + 1 # Simplified example

    n_values = [5, 10, 15, 20, 30, 40]
    results = []

    for n in n_values:
        formula = generate_boolean_formula(n)
        rank = quantum_group_representation(formula)
        depth = dpll_proof_tree_depth(rank)

        if rank > n**1.5 or depth < rank:
            return {
                "metric_name": "DPLL Proof Tree Depth",
                "metric_value": depth,
                "instances_tested": 1,
                "n_max": n,
                "conjecture_holds": False,
                "counterexample": f"Formula: {formula}, Rank: {rank}, Depth: {depth}"
```

```

    }

    results.append({
        "n": n,
        "rank": rank,
        "depth": depth
    })

mean_rank = sum(result["rank"] for result in results) / len(results)
mean_depth = sum(result["depth"] for result in results) / len(results)

return {
    "metric_name": "DPLL Proof Tree Depth",
    "metric_value": mean_depth,
    "instances_tested": len(results),
    "n_max": max(result["n"] for result in results),
    "conjecture_holds": all(depth >= rank for result in results for depth, rank in zip([result["depth"], result["rank"]]))
}

if __name__ == "__main__":
    import sys
    seeds = list(map(int, sys.argv[1:])) or [2**i - 1 for i in range(5, 8)] # Default to first 3

    results = []

    for seed in seeds:
        result = run_trial(seed)
        print(f"TRIAL: {result}")
        results.append(result)

    mean_metric_value = sum(result["metric_value"] for result in results) / len(results)
    support_fraction = sum(1 for result in results if result["conjecture_holds"]) / len(results)

    if all(result["conjecture_holds"] for result in results):
        print(f"RESULT: SUPPORTED mean={mean_metric_value} std=0.0 support_fraction={support_fraction}")
    elif support_fraction >= 0.8:
        print(f"RESULT: SUPPORTED mean={mean_metric_value} std=0.0 support_fraction={support_fraction}")
    else:
        first_failing_seed = next(seed for seed, result in zip(seeds, results) if not result["conjecture_holds"])
        print(f"RESULT: FALSIFIED counterexample=\"First failing seed\" first_failing_seed={first_failing_seed}")

```

6. Per-seed results

(no seed-level data — test crashed before producing TRIAL: lines)

7. Test stdout (last 2KB)

```

'Formula: 11111110101011011111001100000011, Rank: 32, Depth: 33'
TRIAL: {'metric_name': 'DPLL Proof Tree Depth', 'metric_value': 33, 'instances_tested': 1, 'n_max': 5}
TRIAL: {'metric_name': 'DPLL Proof Tree Depth', 'metric_value': 33, 'instances_tested': 1, 'n_max': 5}
TRIAL: {'metric_name': 'DPLL Proof Tree Depth', 'metric_value': 33, 'instances_tested': 1, 'n_max': 5}
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TRIAL: {'metric_name': 'DPLL Proof Tree Depth', 'metric_value': 33, 'instances_tested': 1, 'n_max': 5}
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TRIAL: {'metric_name': 'DPLL Proof Tree Depth', 'metric_value': 33, 'instances_tested': 1, 'n_max': 5}

```

```

TRIAL: {'metric_name': 'DPLL Proof Tree Depth', 'metric_value': 33, 'instances_tested': 1, 'n_max': 5}
TRIAL: {'metric_name': 'DPLL Proof Tree Depth', 'metric_value': 33, 'instances_tested': 1, 'n_max': 5}
TRIAL: {

```

8. Critic adversarial review

Critic verdict: CHALLENGE

Critic reasoning:

The test code uses a trivial placeholder for the quantum group representation logic, returning the length of the formula instead of computing an actual quantum group representation. This is likely to be incorrect and does not scale with n , as it does not reflect the complexity of the problem.

9. Final verdict & safety rail

Verdict: INCONCLUSIVE

Reasoning:

Safety rail: critic_challenge_falsified | original: The test results indicate that for at least one instance (Formula: 00111011011010101011100101001001, Rank: 32, Depth: 33), the minimal rank of a quant | next: Re-evaluate the test code to ensure it accurately computes the quantum group representations and their ranks. If the critic's challenge is valid, further research is needed to address the discrepancy between the computed ranks and DPLL proof tree depths.

11. Audit log (LLM calls)

Total LLM calls: 10

#	Phase	Provider	Model	Tokens in	out	Latency (ms)
1	propose	ollama_remote	glm4:latest	0	0	15020
2	preregistration	ollama_remote	glm4:latest	0	0	9152
3	novelty	ollama_remote	glm4:latest	0	0	12155
4	novelty	ollama_remote	glm4:latest	0	0	8837
5	test_gen	ollama_remote	qwen2.5-coder:7b	0	0	16665
6	test_gen	ollama_remote	qwen2.5-coder:7b	0	0	9163
7	test_gen	ollama_remote	qwen2.5-coder:7b	0	0	8727
8	test_gen	ollama_remote	qwen2.5-coder:7b	0	0	19989
9	critic	ollama_remote	glm4:latest	0	0	47836
10	judge	ollama_remote	glm4:latest	0	0	9792

Totals: 0 input tokens, 0 output tokens, ~\$0.0000 cost, 157337 ms total latency. Provider mix: {'ollama_remote': 10}

(full prompt+response transcripts available in `research/audit/03e72474f51b.jsonl`)

12. Reproducibility

To reproduce this cycle:

1. Apply the test harness in section 5.1 with seeds 11,23,37,53,71
2. The Python sandbox is pure-stdlib; any Python 3.11+ should suffice
3. The full LLM transcripts are in `research/audit/03e72474f51b.jsonl` for verification of provenance

4. The pre-registration hash in section 2 commits to the test design before execution; tampering would be detectable.

Sandbox file: `research/pvsnp_sandbox/test_*.py` (per-cycle)

Replay tarball: `research/replay/03e72474f51b.tar.gz` (if generated)